Does starting age predict who stays? Competition debut and long-term participation in youth track and field

Atle G. Guttormsen[1,*]
ORCID: 0000-0003-0188-8462

[1]School of Economics and Business, Norwegian University of Life Sciences (NMBU), Ås, Norway

*Corresponding author:
Atle G. Guttormsen
School of Economics and Business
Norwegian University of Life Sciences (NMBU)
P.O. Box 5003, 1432 Ås, Norway
E-mail: atle.guttormsen@nmbu.no

Running head: Debut age and retention in youth athletics

Word count: ∼4 100 (main text, excluding abstract, references, tables, and figure legends)

Abstract

Whether starting competitive sport early harms or protects long-term participation is contested: the early-specialisation literature cautions against early entry, while a large prospective dropout study found that dropouts had started *later*. No population study has tested the question in an individual sport. Using the Norwegian Athletics Federation's competition registry, which holds results electronically from 2011, we followed every athlete born 2001–2005 whose first registered competition occurred at ages 10–16 (n = 17 496; 50 % female) to the end of 2026; 546 (3.1 %) were active as seniors (age $\geq$20). Senior conversion was U-shaped in debut age (non-linearity LR test $p < 0.001$): relative to debut at age 10 (3.8 %), adjusted odds were lowest for debuts at 11–14 (OR 0.53–0.75) and highest for debut at 16 (6.6 %; OR 1.73, 95 % CI 1.15–2.59), with a correspondingly null linear trend (OR 0.99 per year, 95 % CI 0.94–1.04). Measured over equal career lengths, however, later starters persisted far less: 20.4 % of those who started at 10 still competed on at least two days in a season four or more years after debut, against 4.5–6.6 % of those who started at 14–16 (OR 0.70 per year of later debut, 95 % CI 0.68–0.72). The fixed-age U combines these faster exits with late entrants' shorter distance to the outcome age. Among athletes competing in the age-17 season, earlier debut appeared advantageous (OR 0.85 per year of later debut, 95 % CI 0.80–0.90), but little association remained once current competition volume was controlled (OR 0.99, 95 % CI 0.92–1.05). Early competitive entry did not undermine retention: per year of career, early entrants persisted longest, in line with the football finding. Among older adolescents, current engagement mattered more than entry history — and every entry age produced senior athletes.

Keywords: youth sport; dropout; starting age; competition entry age; athlete retention; registry study; track and field

1 Introduction

When should children enter competitive sport? The question sits at the centre of the early-specialisation debate, and the two main positions make opposite predictions about participation. One line of argument, prominent in consensus statements, cautions that early competitive engagement risks burnout and withdrawal [1]; on this view, later entry should protect retention. The opposite prediction comes from the retention literature itself: in a large prospective dropout study among 1 762 Dutch football players, those who dropped out had *started organised football later* than those who stayed, and the authors recommended recruiting from early childhood [2] — though its exposure, joining organised football, is not the same construct as the registered-competition debut studied here, a gap we keep explicit throughout. Direct tests are scarce. Markers of early specialisation failed to predict actual dropout in the one study that measured both [3], systematic reviews of youth sport dropout identify motivational and social factors rather than starting age [4, 5], and the retrospective expertise literature — which finds that eventual elites often entered their main sport later [6, 7] — samples on success and therefore cannot speak to population-level retention. No study has followed a complete national entry population of an individual sport and asked what age at entry implies for who is still competing a decade later.

The Norwegian athletics registry permits exactly that design. Every sanctioned result in Norway is registered, the registry holds results electronically from 2011, and competition begins at age 10. For athletes born in 2001 or later, the competitive career is therefore observed from its first registered result, for everyone who ever entered, not only for those who later succeeded — although coverage in 2011, the first electronic season, is thinner than in later years. Three other studies of the same registry frame the present one. Relative-age selection was found to be fully established at the entry stage and stable thereafter, locating the system's selection pressure at the point of entry rather than in adolescent talent identification [8]. A 14-year cohort study found that competition volume at ages 13–14 predicted senior retention somewhat better than performance did, and that exit is preceded by behavioural withdrawal [9]. A companion study of performance trajectories found that progression predicts retention beyond volume — but that, here too, the *recent* window dominates the earlier one [10]. Against that background, entry age

is the natural next exposure: it is the first *registered* behavioural fact the registry holds about an athlete — prior training and other sports are invisible to it.

We pre-specified three questions. **H1:** among all athletes whose first registered competition fell at ages 10–16, is debut age associated with being active as a senior? Read onto this entry-timing margin, the two positions above suggest opposite directions, so H1 is two-sided, a U-shape is plausible, the categorical contrast is the primary display, and a likelihood-ratio test compares the categorical and linear codings. **H2:** once started, do later debutants exit faster, as the football findings suggest [2]? **H3:** among athletes still active at age 17, does entry history add anything beyond current engagement — or does the registry repeat its now-familiar lesson that current behaviour dominates history [9, 10]? Interpretation rules were fixed in advance: the exposure is *competition debut*, not the start of sport participation or training, and selection into late debut (already-interested adolescents choosing athletics) is expected and addressed in the design and interpretation. The analysis plan was committed to version control before any exposure–outcome model was estimated; the study was not externally preregistered.

2 Materials and methods

2.1 Data source and design

The Norwegian Athletics Federation's results database registers every sanctioned competition result in Norway and Norwegian athletes' results abroad. We used a frozen extract of 29 October 2026 (results dated on or before that day). Only participation fields were extracted — athlete, date, meet, club, approval status; no performance data of any kind were used at any age, in keeping with the purpose of the Norwegian children's sport regulations. Age is calendar-year age (competition year minus birth year), the basis of Norwegian age classes. Reporting follows STROBE [11].

2.2 Sample

All athletes born 2001–2008 entered the extraction (n = 33 666, with 675 707 result rows; 1 526 non-approved rows and 116 rows below age 10 were excluded). The 28 339 athletes with an

approved result at age 10 or older constitute the debutant population; 865 without registered sex (predominantly brief, early participants) were excluded, and of the remaining 27 474, 731 (2.7 %) whose debut fell at age 17 or later were set aside as a sensitivity category. Of the remaining 26 743 debutants, the primary sample comprises the 17 496 born 2001–2005 with debut at ages 10–16 — cohorts whose follow-up extends past age 20 for everyone — of whom 50.1 % were female and 546 (3.1 %) met the senior-activity outcome; the other 9 247, born 2006–2008, contribute to descriptive analyses of entry patterns only. Because non-approved rows and sub-age-10 rows were excluded before debut assignment, 49 athletes in the primary sample (0.3 %) had earlier excluded rows, which would change the debut age of 33 of them. The full sample flow is given in Supplementary Table S1.

2.3 Exposure, outcomes, and covariates

The exposure is *debut age*: calendar-year age at the athlete's first approved registered result (10–16), treated categorically (reference: debut at 10) and as a linear trend. The primary outcome, identical to the companion studies [9, 10], is *active senior*: at least two approved results in the same calendar year at age 20 or older. Secondary outcomes were at least one result at age $\geq$20, activity at age $\geq$17, and retention k years after debut (k = 1–5), defined as at least one approved result in the calendar year debut + k (point prevalence), computed with per-k denominators over athlete-years fully observable by 2025. Because the senior outcome is defined at a fixed age, later entrants have fewer years to cover before it; we therefore added an outcome measured at equal career length: at least two competition days in a single season four or more years after the debut season (five and six years as sensitivity analyses), each among athletes for whom that season was fully observable by 2025. Covariates were sex, birth cohort, baseline club (modal club in the debut season) with the logarithm of a club-size proxy (study athletes per club), birth quarter where an exact birth date existed, debut-season competition days, and a COVID-19 flag for debuts in 2020–2021. Covariate sets per hypothesis: the H1 and equal-career-length models adjust for sex, cohort, and club size, with the COVID-19 flag and debut-season days added in robustness variants; the landmark models use sex and cohort, with current competition days added in the second specification. Region could not be adjusted for: the registry records

the competition district for under 1 % of meets. Full coefficient tables are available with the analysis code.

2.4 Statistical analysis

H1 was estimated by logistic regression in the primary sample, with debut age categorical and linear, adjusted for sex, cohort, and club size, with club-clustered standard errors as a sensitivity. H2 was addressed on both clocks without imposing a survival model. On the *duration* clock — the comparison that aligns athletes at the same career stage — we report the proportion still competing k years after their own debut, by debut age, and model the equal-career-length outcome with the H1 covariates. On the *age* clock we report, for the cohorts born 2001–2004 (followed to age 21 for everyone), the share of each debut group with a registered result at that age or later. The analysis plan specified an age-scale Cox model with delayed entry at debut and delayed-entry Kaplan–Meier curves [12]; both were replaced, because exit risk depends on time since debut: the proportional-hazards assumption then fails by construction, and the delayed-entry product-limit estimator multiplies the first-season exits of different entry ages into a single curve that no cohort experiences. Age equals debut age plus career duration, so neither clock isolates an effect of entry age, and both are reported descriptively. The 2026 season is observed to 29 October, which slightly shortens the senior window of the youngest cohort — the cohort covariate and the cohort-split robustness address this. H3 was a landmark analysis [13] among athletes with at least one approved result *in the age-17 season itself* — not merely any result at age 17 or later, which would let future senior activity qualify athletes into the landmark and thereby condition on the outcome's own future. Senior activity was regressed on debut age with sex and cohort, with and without current engagement (competition days at ages 16–17, standardised); the landmark sits at the end of the age-17 season, all predictors are measured by then, and the outcome window begins at age 20. Sex-specific conversion and a sex $\times$ debut-age interaction were also examined. Pre-specified robustness analyses excluded COVID-19 debuts, varied the outcome, added debut-season volume, split the cohorts (2001–2003 vs. 2004–2005), and included debut at 17+ as a category; we added analyses excluding the 2001 cohort, whose debut-at-10 season (2011) has the thinnest coverage, and including

athletes without registered sex as a separate category. The pre-specified debut age × birth-quarter interaction is not reported: exact birth dates exist for 98.5 % of the seniors but 51.9 % of the other athletes, and coverage rises steeply with debut age (Table 1), so the subsample is selected on both outcome and exposure. Analyses used Python 3.13.7 with pandas 2.3.3, numpy 2.4.1, statsmodels 0.14.6 and lifelines 0.30.3.

2.5 Ethics and transparency

The data are competition results registered by the Norwegian Athletics Federation: results from age 13 are publicly listed, while records for the 10–12 age classes are registered but not publicly released by the federation, and for those ages only participation itself (whether and when an athlete competed) was used. No individual is identified and no performance data were used. As for the companion registry studies, the project falls outside the scope of the Norwegian Health Research Act, and an assessment from Sikt was obtained. The analysis plan, hypotheses, estimands, and interpretation rules were committed to version control before estimation; code and aggregate outputs are available, and athlete-level files are regenerable from the registry.

3 Results

3.1 Who debuts when

Debut at 10, the first eligible age, was the single most common entry point, with a share rising from 32.6 % of debutants born 2001 to 43.9 % of those born 2008 — a rise exaggerated at both ends, since the 2001 cohort's debut-at-10 season was 2011, the first and thinnest electronic season, and the 2008 cohort lost most debuts at 12 to the cancelled 2020 season (9.3 % debuted at 12, against 17.1–17.5 % in the two preceding cohorts). Late entry remained substantial: roughly one debutant in six entered at 14 or later (Figure 1). Table 1 describes the three debut groups. Late debutants were more often male (female share 52.2 % for debut at 10–11 against 42.0 % for 14–16). The typical debut was brief in every group: the median debutant competed on a single day in the debut season, and median career length was two seasons for debut at 10–11 and one season otherwise. Entry intensity was thus flat across debut ages — debut age is

not a proxy for dose at entry. Exact birth dates — needed for birth-quarter analyses — existed for 38.6 % of early, 63.8 % of mid, and 88.5 % of late debutants, the registry's known pattern of missingness concentrating among brief early participants [8]; among those with known dates, the familiar first-quarter overrepresentation was present at every single debut age (Q1 25–31 % against Q4 19–22 %).

Table 1: Characteristics of the primary sample (born 2001–2005, debut at ages 10–16), by debut group.

Characteristic	Debut 10–11	Debut 12–13	Debut 14–16
n	10 242	4 220	3 034
Female, %	52.2	50.8	42.0
Exact birth date available, %	38.6	63.8	88.5
Born Q1, % of those with birth date	29.2	26.1	27.1
Born Q4, % of those with birth date	19.9	21.9	19.8
Competition days in debut season, median (IQR)	1 (1–3)	1 (1–3)	1 (1–2)
Career length, seasons, median (IQR)	2 (1–4)	1 (1–3)	1 (1–2)
Debut season 2020–2021, %	0.0	0.0	6.3

By construction, cohorts born 2001–2005 could only experience a 2020–2021 debut at ages 15 and 16.

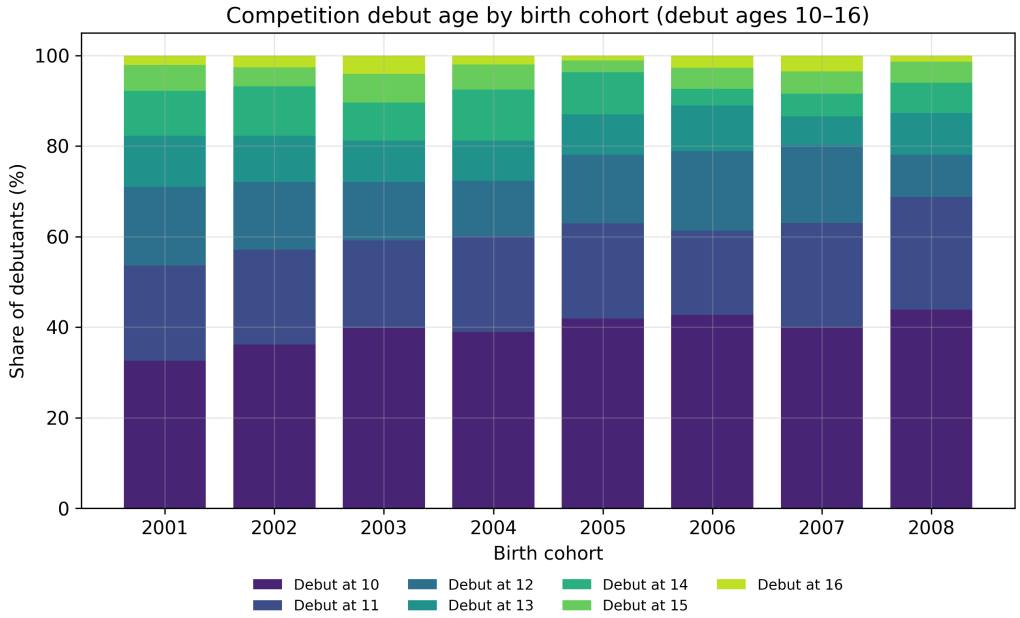


Figure 1: Competition debut age by birth cohort (all debutants born 2001–2008, debut ages 10–16).

3.2 Senior conversion is U-shaped in debut age (H1)

Of the 17 496 debutants, 546 (3.1 %) were active as seniors. Conversion was not monotone in debut age (Figure 2): it was 3.8 % for debut at 10 (253/6 626), fell to 2.4 % at 11, 2.1 % at 12, 2.9 % at 13 and 2.4 % at 14, and rose to 3.9 % at 15 and 6.6 % at 16 (28/424). A likelihood-ratio test rejected the linear coding in favour of the categorical one (LR = 43.2, df = 5, $p = 3.3 \times 10^{-8}$): the association is non-linear. Adjusted for sex, cohort, and club size (Table 2), debuts at 11, 12 and 14 had significantly lower odds than debut at 10 (OR 0.53–0.63), debut at 13 directionally lower odds (OR 0.75, 95 % CI 0.55–1.02), debut at 15 the same odds (OR 1.00), and debut at 16 markedly higher odds (OR 1.73, 95 % CI 1.15–2.59, resting on 28 events). The linear trend across 10–16 was correspondingly null (OR 0.99 per year, 95 % CI 0.94–1.04) — a flat average concealing a U. Because the outcome is defined at a fixed age, these rates compare entrants who had between four (debut at 16) and ten (debut at 10) seasons to cover before the outcome window; Section 3.3 measures persistence at equal career length. Conversion was higher for boys than girls at every debut age (4.3 vs. 3.4 % at debut 10; 7.1 vs. 5.7 % at debut 16; Supplementary Table S3), and the late-entry channel is male-tilted in composition (42.0 %

female at debut 14–16); the sex × debut-age interaction was directionally negative but not significant (OR 0.92 per year, 95 % CI 0.82–1.02), so the U describes both sexes.

Table 2: Senior activity (age ≥20) and competing at equal career length, by debut age (n = 17 496; 546 senior events, 2 312 equal-career-length events). % = raw proportion; odds ratios relative to debut at age 10, adjusted for sex, birth cohort, and club size. Competing ≥4 years after debut: at least two competition days in a single season four or more years after the debut season. A likelihood-ratio test favours the categorical over the linear coding for the senior outcome (LR = 43.2, df = 5, $p < 0.001$); the debut-16 senior estimate rests on 28 events.

		Active senior (age ≥20)		Competing ≥4 years after debut	
Debut age	n	%	OR (95 % CI)	%	OR (95 % CI)
10	6 626	3.8	1.00 (ref.)	20.4	1.00 (ref.)
11	3 616	2.4	0.63 (0.49–0.81)	12.4	0.56 (0.50–0.63)
12	2 529	2.1	0.53 (0.39–0.71)	8.5	0.37 (0.31–0.43)
13	1 691	2.9	0.75 (0.55–1.02)	8.7	0.37 (0.31–0.44)
14	1 747	2.4	0.62 (0.44–0.86)	4.5	0.19 (0.15–0.24)
15	863	3.9	1.00 (0.70–1.45)	4.9	0.21 (0.15–0.29)
16	424	6.6	1.73 (1.15–2.59)	6.6	0.29 (0.20–0.43)
Linear trend, per +1 year			0.99 (0.94–1.04)		0.70 (0.68–0.72)

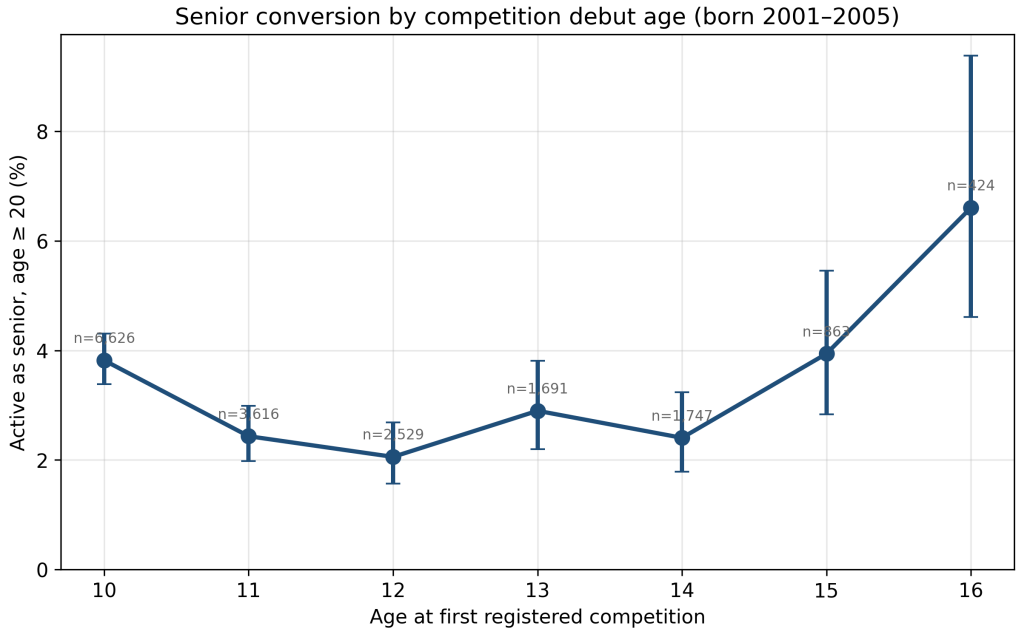


Figure 2: Senior conversion by competition debut age, with Wilson 95 % confidence intervals (born 2001–2005).

3.3 Later starters leave faster — on their own clock (H2)

Aligned on time since debut (Table 3; Figure 3), early entrants were far more durable. One year after the debut season, 51 % of those who debuted at 10 were still competing, against 34–40 % of those who debuted at 11–14, 26 % at 15, and 35 % at 16. Five years after debut the gap had widened to 17 % (debut 10) and 10 % (debut 11) against 3–7 % (debuts 12–16). Denominators are fully observable athlete-years and shrink only for debut at 16 (424 at $k = 1$ to 389 at $k = 5$); Wilson intervals for every cell are given in Supplementary Table S2. The equal-career-length outcome gives the same picture in one model (Table 2): 20.4 % of those who debuted at 10 competed on at least two days in a season four or more years later, against 12.4 % at 11, 8.5–8.7 % at 12–13 and 4.5–6.6 % at 14–16, an adjusted OR of 0.70 per year of later debut (95 % CI 0.68–0.72), essentially unchanged at five (0.69) and six years (0.66; Supplementary Table S4). The fixed-age senior outcome therefore combines two opposing forces: later entrants exit faster per year of career but have fewer years to cover before age 20, and the null linear trend in H1 is these forces cancelling. On the age clock (Figure 4), the late group had far more athletes still competing at 16 (31.0 % with a result at 16 or later, against 14.0 and 13.7 % in the

earlier groups) — partly by construction, since it includes the athletes who debuted at 16 — and the groups were close from about 19: at 20, 4.2, 3.3 and 4.5 % of the three groups were still competing, and at 21, 3.4, 2.7 and 3.1 %. Early entry thus did not leave clearly more athletes competing at adult ages; it bought a longer career from an earlier start.

Table 3: Proportion still competing k years after the debut season (at least one approved result in the calendar year debut + k), by debut age (born 2001–2005; athlete-years fully observable by 2025). Denominators equal the stated n except for debut at 16, where they shrink from 424 ($k \leq 4$) to 389 (k = 5); Wilson 95 % intervals for every cell are given in Supplementary Table S2.

Debut age	n	k=1	k=2	k=3	k=4	k=5
10	6 626	0.51	0.38	0.29	0.22	0.17
11	3 616	0.40	0.27	0.19	0.13	0.10
12	2 529	0.37	0.23	0.14	0.09	0.07
13	1 691	0.38	0.23	0.15	0.10	0.07
14	1 747	0.34	0.13	0.08	0.05	0.03
15	863	0.26	0.13	0.08	0.05	0.04
16	424	0.35	0.21	0.11	0.07	0.06

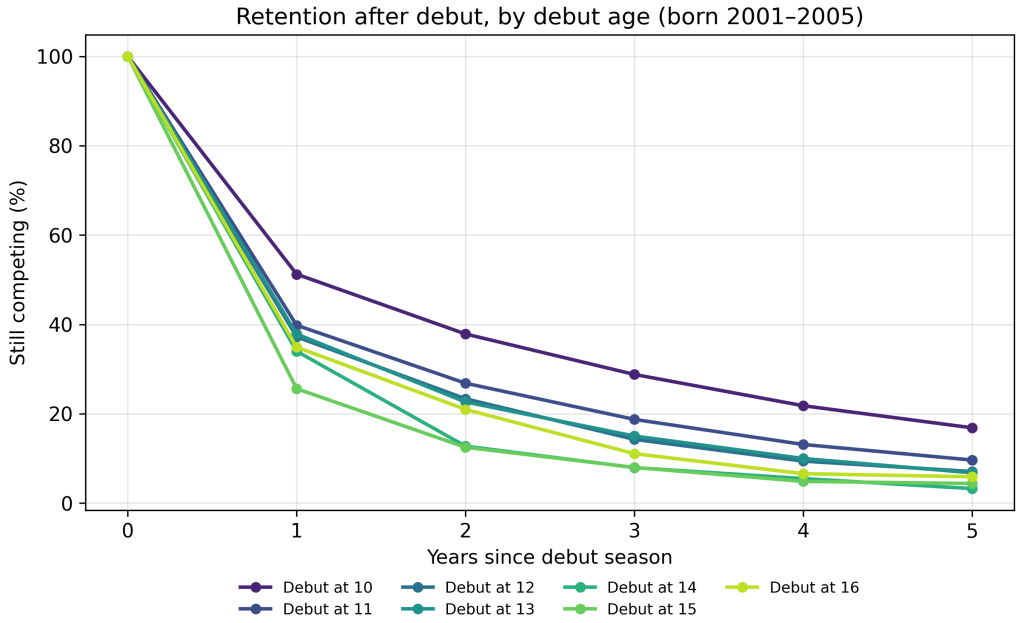


Figure 3: Retention k years after the debut season, by debut age (born 2001–2005).

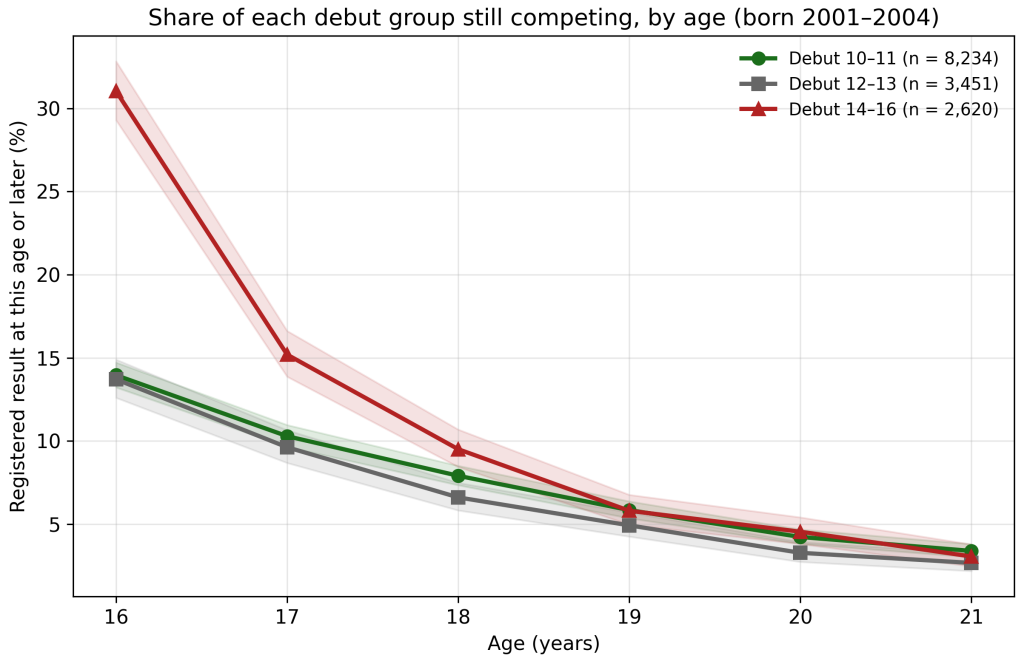


Figure 4: Share of each debut group with a registered result at that age or later, cohorts born 2001–2004 (followed to age 21 for everyone), with Wilson 95 % confidence bands. At 16 the late group includes the athletes who debuted at that age.

3.4 Among survivors, entry history adds little beyond current engagement (H3)

Of the 17 496 debutants, 1 637 (9.4 %) competed in the age-17 season itself; 472 of these (28.8 %) became active seniors, while 74 seniors had a fully idle age-17 season before returning — which is exactly why the landmark must be defined on the season itself rather than on any activity at age 17 or later. In this landmark sample, earlier debut appeared advantageous: OR 0.85 per year of later debut (95 % CI 0.80–0.90). Adding current engagement — competition days at ages 16–17 — brought the association close to null (OR 0.99, 95 % CI 0.92–1.05). Conditional on competing at 17 and on how much the athlete competes now, when they started carries little additional predictive information. Two readings must nevertheless be kept apart: current volume is plausibly a *mediator* of entry timing, and conditioning on survival to the landmark invites index-event selection, so the result licenses no claim that entry history is causally inert — only that it adds little to prediction at the landmark. This replicates, for entry history, what the companion studies found for early competition volume [9] and early performance [10]: in this registry, recent behaviour dominates history.

3.5 Robustness

None of the variations (Table 4) moved the linear trend far from null. Excluding COVID-era debuts gave 0.95 (95 % CI 0.90–1.01); excluding the 2001 cohort, whose debut-at-10 season fell in the thinly covered 2011, gave 0.99; including the 549 athletes without registered sex as a separate category gave 0.99, with debut-10 conversion 3.7 instead of 3.8 %; the one-result senior outcome gave 1.00; activity at 17 among debuts at 10–14 gave 0.97; conditioning on debut-season volume gave a borderline 0.94 (95 % CI 0.89–1.00); and the cohort halves agreed (0.98 and 0.99). Athletes debuting at 17–19 — outside the primary range, and excluding the 165 athletes whose first registered result came at age 20 or later and could therefore coincide with the outcome itself — converted at distinctly higher rates (OR 3.36 vs. debut at 10, 95 % CI 2.42–4.67; within the late group, 193 debuted at 17, 153 at 18 and 56 at 19), an extension of the U's right arm that is close to mechanical: a debut at 19 is a single season from the outcome

window.

Table 4: Robustness analyses. Estimates are for debut age (per +1 year, linear) unless stated; outcome is senior activity unless stated.

Analysis	n	Events	OR (95 % CI)
Primary linear trend	17 496	546	0.99 (0.94–1.04)
Excluding debuts in 2020–2021	17 305	530	0.95 (0.90–1.01)
Excluding the 2001 cohort (debut at 10 in 2011)	14 077	434	0.99 (0.94–1.05)
Including athletes without registered sex	18 045	547	0.99 (0.94–1.04)
Outcome: $\geq$1 result at age $\geq$20	17 496	694	1.00 (0.96–1.05)
Outcome: any result at age $\geq$17 (debuts 10–14)	16 209	1 621	0.97 (0.93–1.00)
Adding debut-season competition days	17 496	546	0.94 (0.89–1.00)
Cohorts born 2001–2003	10 667	349	0.98 (0.92–1.05)
Cohorts born 2004–2005	6 829	197	0.99 (0.91–1.08)
Debut at 17–19 vs. debut at 10 (categorical)	17 898	595	3.36 (2.42–4.67)
Adding the COVID-19 debut flag (linear trend)	17 496	546	0.95 (0.90–1.01)
Club-clustered robust SE (trend)	17 496	546	0.99 (0.92–1.06)

4 Discussion

4.1 Principal findings

In the complete entry population of Norwegian youth track and field, age at first competition carried no simple message about who would still be competing as an adult. Senior conversion was U-shaped — highest for the earliest (10) and the latest (16, and 17+) entrants, lowest in between — with a null linear trend. Measured from their own start, later entrants churned much faster, half of the ten-year-old debutants surviving the first year against roughly a quarter to a third of those debuting at 14–16, and at equal career length the odds of still competing were 30 % lower per year of later debut. The fixed-age U is the sum of these faster exits and the shorter distance late entrants have to the outcome age. Among athletes competing in the age-17 season, entry history added little predictive information once current competition volume was

known.

4.2 What the entry-timing margin can and cannot test

Neither of the positions that motivated the study was formulated about this exposure, and the revision of both must respect that. The specialisation cautions concern specialised, high-volume, year-round involvement [1] — not whether a ten-year-old appears once in a sanctioned meet (the median debutant competed on a single day); the football finding concerns joining organised sport [2], whereas a "late debutant" here may have trained for years before a first meet. What this study tests directly is *entry timing into registered competition* — an administrable, policy-relevant variable federations actually control records for — and it comments on the specialisation debate only indirectly. On that margin, early entry carried no retention penalty: early debutants converted better than the 11–14 entrants at the fixed age and were by far the most durable on the duration clock. The high fixed-age conversion of those who entered at 15–16, and especially 17–19, largely reflects how close their debut lies to the outcome window; at equal career length they persisted least, as the football study found for later joiners [2].

Beyond that mechanical distance, the two arms plausibly reflect selection, with different candidate mechanisms. On the right arm, entry at 15 and later is plausibly a decision adolescents make for themselves, and those who make it are a small, motivated, male-tilted group whose careers, when they fail, fail fast. On the left arm, several non-exclusive mechanisms could make debut at 10 beat 11–14: families with the strongest athletics attachment enrol at the first eligible age; mid-childhood entrants may be samplers arriving from other activities with weaker athletics identity; and club pathways built around training groups formed at 10 may serve later joiners less well. We present the parental-decision reading of early entry as interpretation, not evidence. And the fullest honesty is this: the entire U may be compositional — debut age may carry no behavioural effect at all, only information about who enters when. The registry cannot see motivation, prior training, or participation in other sports, and competition debut is not the start of sport involvement.

4.3 The first season is the universal filter

The dominant feature of every debut group is the first-year cliff: the median debutant competed on one day, and between half and three quarters of each entry cohort never established themselves. This parallels the registry's relative-age structure, whose selection is already complete at entry rather than produced in adolescent talent identification [8], and it extends that point from birth-date composition to participation as such: most of the selection happens at or soon after entry. Whatever determines long-term engagement, most of its action is over within a season or two of arrival, at every entry age.

4.4 History versus current engagement, third instalment

The landmark result completes a pattern now replicated three times in this registry with three different exposures. Early competition volume predicts senior retention, but later volume dominates it [9]. Early performance progression predicts retention, but through the most recent window [10]. And entry age predicts little among 17-year-old survivors once current volume is in the model. For talent systems, the registry keeps returning one lesson: *as a predictor*, what an athlete is doing now outperforms any summary of how they got there. The claim is strictly predictive. Current engagement is plausibly a mediator of entry history, and the landmark conditions on survival to 17, so the attenuation of the debut coefficient is what a mediated exposure would show even if entry timing mattered causally. What follows for practice is correspondingly modest: at the landmark, entry-age heuristics add little to what current participation already says, in either direction.

4.5 Strengths and limitations

The strengths are completeness and design: a national registry that observes every registered debut from the first eligible age, for everyone who ever entered (not only later successes, the known weakness of retrospective expertise samples [6, 7]), comparisons on both the age and the duration clock, a plan committed before estimation, and internal replication across cohort halves. The limitations are equally structural. Competition debut is not sport start: children

may train for years before their first meet, and late debutants may arrive from other sports with long athletic histories, so the exposure is entry into *registered competition*. Exit from the registry is not exit from sport. Selection into debut age is unmeasured and surely strong; no causal reading is offered. Birth dates are missing for most brief early participants — 98.5 % of the seniors but 51.9 % of the other athletes have one — so birth-quarter shares are descriptive and no birth-quarter interaction is reported. Coverage in 2011, the first electronic season, is thinner than in later years, so some debuts at 10 in the 2001 cohort may be missing; excluding that cohort did not change H1. Registered debut also reflects registration practice — which meets submit results for the youngest classes, and how clubs handle them — so cohort trends in debut age (Figure 1) may be partly organisational rather than behavioural. Selection into debut age plausibly tracks geography, club culture, family resources, and socioeconomic access, none of which the registry measures, and the club-size proxy is defined within the study sample. Retention is not attainment: no performance data were used, so nothing here says where elite performers come from — the retrospective expertise literature cannot speak to retention, and this study cannot speak to attainment. The senior outcome is rare (3.1 %), so even twofold relative differences are small in absolute terms. Finally, age equals debut age plus career duration, so no comparison separates an effect of entry age from those of age and career stage; the analysis plan's age-scale Cox model was replaced for this reason, and all H2 comparisons are descriptive.

5 Perspective

For federations and clubs, these population figures provide three descriptive planning parameters. First, early entry carries no retention penalty, and no age is too late to start: every entry age from 10 to 19 produced senior athletes. The latest entrants' high per-head conversion at a fixed adult age reflects their short distance to that age and *self-selection*, not an advantage of entering late — at equal career length, later entrants persisted less. Adolescents recruited into entry who would not have come on their own cannot be assumed to convert at the observed 6.6 %, and the late channel as observed is male-tilted, which matters for what a recruitment drive would actually reach. Second, the decisive window is the first season or two after entry,

whenever entry happens: between half and three quarters of every entry cohort resolves within two seasons, so retention effort belongs at the entry transition rather than in managing entry timing — a conclusion that does not depend on the selection caveat. Third, by late adolescence entry history adds little: among those competing at 17, current engagement carries most of the predictive information. All three statements concern participation in registered athletics, not sport participation or physical activity in general.

Declarations

Funding. The author received no funding for this work.

Competing interests. The author declares no competing interests. The author maintains an open, non-commercial statistics platform built on the federation's public results database.

Ethics. The study analyses competition results registered by the Norwegian Athletics Federation (publicly listed from age 13; registered but not publicly released for the 10–12 age classes) and falls outside the scope of the Norwegian Health Research Act; an assessment from Sikt was obtained. No individual athlete is identified. No performance data were used, and no analysis below age 13 goes beyond participation itself, in keeping with the Norwegian children's sport regulations.

Data availability. The underlying data are the Norwegian Athletics Federation's competition results; records for the 10–12 age classes are not publicly released and cannot be shared at the individual level. Aggregate outputs and all analysis code are available from the author; athlete-level files are regenerable from the registry with the analysis code.

Author contributions. AGG conceived the study, performed all analyses, and wrote the manuscript.

Declaration of AI use. The author used a large language model (Claude, Anthropic) for programming assistance and manuscript drafting support. The author reviewed, verified, and edited all content and takes full responsibility for the integrity of the work.

Acknowledgements. The author thanks the Norwegian Athletics Federation, whose systematic registration of competition results makes population research of this kind possible.

Supplementary material

Table 5: *

Supplementary Table S1. Sample flow.

Step	n
Athletes born 2001–2008 in the registry extract	33 666
Result rows extracted (2011 – 29 Oct 2026)	675 707
excluded: non-approved rows	1 526
excluded: rows below age 10	116
Debutants with an approved result at age $\geq$10	28 339
excluded: missing sex	865
set aside: debut at age $\geq$17 (sensitivity analyses)	731
Debutants at ages 10–16 with known sex	26 743
primary sample: born 2001–2005	17 496
descriptive only: born 2006–2008	9 247
Active seniors in the primary sample	546

Table 6: *

Supplementary Table S2. Retention k years after the debut season: proportion (Wilson 95 % CI). Denominators equal the debut-age n except debut 16 at k = 5 (n = 389).

Debut age	n	k=1	k=2	k=3	k=4	k=5
10	6 626	.512 (.500–.524)	.379 (.367–.390)	.288 (.277–.299)	.218 (.208–.228)	.169 (.160–.178)
11	3 616	.398 (.382–.414)	.269 (.254–.283)	.188 (.175–.201)	.131 (.121–.143)	.097 (.087–.107)
12	2 529	.372 (.353–.391)	.234 (.218–.251)	.143 (.130–.157)	.094 (.083–.106)	.071 (.061–.081)
13	1 691	.378 (.356–.402)	.226 (.207–.247)	.150 (.134–.168)	.100 (.087–.115)	.068 (.057–.081)
14	1 747	.340 (.318–.363)	.128 (.113–.144)	.080 (.068–.093)	.055 (.045–.067)	.033 (.025–.042)
15	863	.256 (.228–.286)	.125 (.105–.149)	.080 (.064–.100)	.049 (.036–.065)	.044 (.032–.060)
16	424	.349 (.305–.396)	.210 (.174–.251)	.111 (.084–.144)	.066 (.046–.094)	.059 (.040–.087)

Table 7: *

Supplementary Table S3. Senior conversion by debut age and sex (born 2001–2005), % (Wilson 95 % CI).

Debut age	Girls, n	Girls, % (CI)	Boys, n	Boys, % (CI)
10	3 485	3.4 (2.8–4.0)	3 141	4.3 (3.6–5.1)
11	1 860	2.2 (1.6–2.9)	1 756	2.7 (2.1–3.6)
12	1 263	1.8 (1.2–2.7)	1 266	2.3 (1.6–3.3)
13	879	2.0 (1.3–3.2)	812	3.8 (2.7–5.4)
14	789	2.0 (1.3–3.3)	958	2.7 (1.9–3.9)
15	326	2.5 (1.2–4.8)	537	4.8 (3.3–7.0)
16	158	5.7 (3.0–10.5)	266	7.1 (4.6–10.9)

Sex × debut-age interaction (linear): OR 0.92 per year for girls relative to boys, 95 % CI 0.82–1.02. Full coefficient tables for the primary models are available with the analysis code.

Table 8: *

Supplementary Table S4. Competing at equal career length: proportion with at least two competition days in a single season at least 4, 5 and 6 years after the debut season, by debut age (n = athletes for whom that season was fully observable by 2025), with the adjusted OR per year of later debut (sex, birth cohort, club size).

Debut age	4 years: % (n)	5 years: % (n)	6 years: % (n)
10	20.4 (6 626)	15.7 (6 626)	12.0 (6 626)
11	12.4 (3 616)	8.8 (3 616)	6.4 (3 616)
12	8.5 (2 529)	6.2 (2 529)	4.0 (2 529)
13	8.7 (1 691)	6.2 (1 691)	4.1 (1 691)
14	4.5 (1 747)	2.9 (1 747)	2.2 (1 747)
15	4.9 (863)	3.8 (863)	2.6 (779)
16	6.6 (424)	5.1 (389)	3.2 (317)
Linear trend, OR per +1 year	0.70 (0.68–0.72)	0.69 (0.66–0.72)	0.66 (0.63–0.69)

References

[1] John P. DiFiori, Holly J. Benjamin, Joel S. Brenner, Andrew Gregory, Neeru Jayanthi, Greg L. Landry, and Anthony Luke. Overuse injuries and burnout in youth sports: a position statement from the American Medical Society for Sports Medicine. *Clinical Journal of Sport Medicine*, 24(1):3–20, 2014.

[2] Nico W. Van Yperen, Laura Jonker, and Jan Verbeek. Predicting dropout from organized football: a prospective 4-year study among adolescent and young adult football players. *Frontiers in Sports and Active Living*, 3:752884, 2022.

[3] Heather K. Larson, Bradley W. Young, Tara-Leigh F. McHugh, and Wendy M. Rodgers. Markers of early specialization and their relationships with burnout and dropout in swimming. *Journal of Sport and Exercise Psychology*, 41(1):46–54, 2019.

[4] Jeff Crane and Viviene Temple. A systematic review of dropout from organized sport among children and youth. *European Physical Education Review*, 21(1):114–131, 2015.

[5] Lars Erik Espedalen and Ørnulf Seippel. Dropout and social inequality: young people's reasons for leaving organized sports. *Annals of Leisure Research*, 27(2):197–214, 2024.

[6] Michael Barth, Arne Güllich, Brooke N. Macnamara, and David Z. Hambrick. Predictors of junior versus senior elite performance are opposite: a systematic review and meta-analysis of participation patterns. *Sports Medicine*, 52(6):1399–1416, 2022.

[7] Arne Güllich, Michael Barth, David Z. Hambrick, and Brooke N. Macnamara. Recent discoveries on the acquisition of the highest levels of human performance. *Science*, 390(6779):eadt7790, 2025.

[8] Atle G. Guttormsen. Dropped out before they ever competed: a population-wide registry analysis of relative age effects in Norwegian track and field from age 10 to 25. *Sports Medicine – Open*, 2026. In press.

[9] Atle G. Guttormsen. Pulling back before dropping out: declining competition participation

precedes exit from Norwegian youth track and field — a 14-year register study. Under review, 2026.

[10] Atle G. Guttormsen. Stagnation before exit: performance progression and retention in youth track and field. Manuscript in preparation, 2026.

[11] Erik von Elm, Douglas G. Altman, Matthias Egger, Stuart J. Pocock, Peter C. Gøtzsche, and Jan P. Vandenbroucke. The Strengthening the Reporting of Observational Studies in Epidemiology (STROBE) statement: guidelines for reporting observational studies. *The Lancet*, 370(9596):1453–1457, 2007.

[12] Edward L. Korn, Barry I. Graubard, and Douglas Midthune. Time-to-event analysis of longitudinal follow-up of a survey: choice of the time-scale. *American Journal of Epidemiology*, 145(1):72–80, 1997.

[13] James R. Anderson, Kevin C. Cain, and Richard D. Gelber. Analysis of survival by tumor response. *Journal of Clinical Oncology*, 1(11):710–719, 1983.